Charotar University of Science and Technology

M.Sc (Advance Organic Chemistry) Semester- II Examination May 2011

OC -708: Group Theory & Spectroscopy

Date: May 03, 2011 Time: 10.00 am to 10.30 am Max. Marks: 20

Part A

1. What is an orthogonal matrix? If two matrices A and B are orthogonal, show that the product AB is also orthogonal. (3)

2. For a linear moiety HF^{2-}, the Raman peak at $675 cm^{-1}$ does not change when H is replaced by D. This peak is due to: (2)
   i. Bending ii. Symmetric stretching iii. Out of plane bending iv. Anti -symmetric stretching.

3. The microwave spectrum of HCl molecule shows two consecutive absorptions at 20.7 and 41.5 cm^{-1}. Obtain the value of B. (3)

4. The presence of two C=O stretching vibrations at 1696 and 1715 cm^{-1} in CCl_4 solution of $C_6H_5COCH_2Cl$ is due to : (2)
   i. field effect. ii. Inductive effect. iii. mesomeric effect.

5. A monochromatic radiation is allowed to pass through an assembly of molecules. The change in the frequency of the scattering radiation is called: (2)
   i. Raman Scattering ii. Rayleigh scattering iii. Tyndall scattering .

6. The XPES of CH_3CH_2OCOCl shows three C(1s) peaks at 290, 286 and 283 eV. Assign these to appropriate carbons. (2)

7. Show that a given element occurs once and only once in a given row or column in a multiplication table, derived from the elements of a group. (3)

8. The IR spectra of benzyl chloride shows a strong peak at 1790 cm^{-1} and a weak one at 1750 cm^{-1}. Explain the nature of these absorptions. (3)

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Note: Character table will be provided on demand

Part B

1(a). Answer any one: (4)

i. Find the atomic orbitals of B required to form σ bonds in BF_3.

ii. Find the atomic orbitals of Br required to form σ bonds in BrF_5.

(b). With reference to appropriate rules, construct the character table for a group consisting of four elements E, C_{2V}, σ_V and σ'_V. Each element is a class by itself. (4)

(c). Obtain the transformation matrix for a clockwise rotation through angle θ about Z axis. (2)

OR

(c). Explain the physical significance of class.

2. Provide the MO description of π bonding in planar trigonal molecule CO_3^{-2}. (10)

OR

2. Obtain the LCAO for σ bonding in planar XeF_4 molecule.

3(a). Derive an expression to calculate the number of rotations /sec for a diatomic molecule at the J^{th} rotational level. (3)

OR

(a). The rotational constant for HCl^{35} is 10.5909 cm^{-1}. Calculate the B for HCl^{37}.

(b). The rotational spectrum of $^{79}Br^{14}F$ shows a series of equidistant lines spaced 0.71433 cm^{-1} apart. Find which transition gives rise to most intense spectral line at 300^0 K. (5)

(c). Derive the expression for the moment of inertia of a symmetrical linear tetra atomic molecule. (2)

4(a). Show that the change in the frequency of a monochromatic radiation due to its passage through a medium may be attributed to the change in the internal energy of the scattering medium. (4)

(b). The CN stretching in m-methyl aniline appears at a higher frequency than that in p-methyl aniline. Explain. (3)

OR

(b). The intensity of Raman Stokes lines is greater than that of the anti Stokes lines. Justify.

(c). The normal modes of vibration in free NO_3^- are classified as $2E' + A_1' + A_2''$. Discuss the IR and Raman activities of these modes. Under C_{2V} symmetry the vibrations of NO_3 are classified as $3A_1 + 2B_1 + B_2$. Explain from the information provided, how many vibrations will be IR active in both mono- dentate and bi- dentate nitrate complexes. (3)

OR

(c). The equilibrium frequency of vibration of I_2 is 215 cm^{-1} and anharmonicity constant is 0.003. What at 300°K is the intensity of the hot band relative to that of the fundamental?

5.(a). How is Photoelectron spectral data employed to identify the interaction between lone pairs? (3)

(b). Calculate the effective charge on N atom in both nitrobenzene and aniline. Predict which one of the two N atoms will give a peak at higher energy. The electro negativities are: (5)

$$\chi_H = 2.1,\ \chi_N = 3.0,\ \chi_{N^+} = 3.3,\ \chi_C = 2.5,\ \chi_O = 3.5,\ \chi_{O^-} = 3.2$$

OR

(b). The values of w_e and $w_e x_e$ are 691.75 and 4.720 cm^{-1} respectively for NBr^{79}. Calculate the bond dissociation energy of NBr^{79} in kcal/mol. Also, obtain the value of v (vibrational level) where the dissociation starts.

(c). Explain Auger effect and X-ray fluorescence. (2)